CHEBYSHEV GREEDY ALGORITHM FOR NONLINEAR PDES

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Abstract. We study the Chebyshev greedy algorithm (CGA) for convex nonlinear partial differential equation energies using shallow ReLU^k ridge functions, selected from finite candidate pools with inexact coefficient solves. The one-step estimate separates four contributions: coverage of the continuous dictionary by the pool, evaluation of the selection score, radial stationarity in the active space, and the accuracy of the coefficient optimization. Under an atomic source condition, directional smoothness, uniformly positive effective selection, and relative error control, this estimate yields an $O(m^{-1})$ energy bound. For exact coefficient minimization, local Hessian transfer relates nonlinear selection to weak orthogonal greedy approximation under relative-score, invariant-neighborhood, and entropy conditions. For the nested spaces generated by the algorithm, an approximate-source projection bound keeps both the source remainder and the observed orthogonal increments. We illustrate how an exact energy expansion and explicit comparison conditions convert projection estimates into nonlinear energy-error bounds. A separate power-type decrease hypothesis gives an explicit projection bound on a finite index range. For standard p-Laplacian energies with $p \geq 2$, an energy exponent α yields natural-distance and Sobolev exponents $\alpha/2$ and α/p. Eight one- and two-dimensional problems illustrate the estimates, including five comparisons with finite elements and random features at prescribed coefficient counts under a common evaluation protocol. For the two quartic auxiliary models, the resulting bounds cover the computed energy errors over the calibration range; continuation with fixed constants tests the power-type decrease condition.

Key words. Chebyshev greedy algorithm; nonlinear partial differential equation; shallow neural network; p-Laplacian; natural distance; finite dictionary; orthogonal greedy approximation

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1. Introduction.

Variational formulations turn a broad class of elliptic boundary-value problems into convex energy minimization. Classical finite elements approximate the admissible space by locally supported polynomials, and the relation between polynomial degree, mesh scale, and degrees of freedom is well understood [3]. Neural and ridge-function methods instead construct global nonlinear trial spaces, so that the approximation problem concerns the selection and reoptimization of atoms rather than the design of a mesh. The deep Ritz method illustrates the reach of variational neural solvers [7]. The present work analyzes a deterministic greedy selection rule for such nonlinear elliptic energies.

Orthogonal greedy approximation (OGA) has a sharp Hilbert-space theory. Under suitable source and entropy conditions it produces optimal approximation rates [14, 9]. Chebyshev-type greedy methods extend the basic descent argument to convex energies in Banach spaces [17, 18], and biorthogonal or inexact variants show how selection and correction errors enter the energy recursion [5]. These results supply the approximation theory on which the analysis below relies. Applying these results to a nonlinear partial differential equation (PDE) approximated using atoms drawn from a finite candidate pool also requires accounting for numerical quadrature, finite-dictionary approximation, and inexact coefficient minimization. The three effects are distinct: quadrature perturbs the evaluated score, a finite pool replaces the continuous supremum by a maximum over

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the pool, and an active-space solver computes a minimizer to a prescribed accuracy.

The dictionary used below consists of shallow ReLU^k ridge functions. Here $\mathrm{ReLU}^k(s) = [s]_+^k$, with $[s]_+ = \max\{s, 0\}$, and the domain is a bounded subset of $\mathbb{R}^d$. Greedy approximation with neural networks has been used successfully for PDEs [13]. A pool that is drawn once and then held fixed during the greedy solve provides a way to formulate weak selection, by which we mean selection of an atom whose score is a prescribed fraction of the best score available in the full dictionary: [19, Theorem 3.1] prove a convex-energy bound under Hilbert smoothness and source/entropy assumptions, while their Theorems 5.1 and 5.2 quantify dictionary sizes that realize weak selection. Their analysis assumes exact coefficient minimization over the active span. The finite-pool inequality below separates the correction error from the score and coverage errors. The universality of such constructions for nonlinear convex variational problems is examined by [2]. Variation spaces and sharp entropy estimates for shallow networks provide the source and width information used in the convergence estimates [15, 16, 10].

The analysis has three parts. First, the one-step descent estimate separates pool coverage, score evaluation, radial stationarity, and coefficient-correction errors. Second, local Hessian transfer connects nonlinear greedy selection with weak OGA in a frozen Hilbert metric under relative-score, invariant-neighborhood, and entropy conditions. The entropy estimate uses the projected, unrenormalized restarted dictionary. Third, we estimate the projection error in the spaces generated by the algorithm. These estimates account for the source remainder and the observed orthogonal increments. For quartic energies, an exact identity converts the projection estimates into nonlinear energy-error bounds. The numerical study evaluates the finite-trajectory estimates on the computed spaces.

For p-growth energies, the natural distance describes the geometry of the energy gap. The nonlinear map

$$V(\xi) = |\xi|^{(p-2)/2}\xi$$

defines the natural distance, as in the finite-element analysis of the p-Laplacian [1, 6, 8]. For the standard p-energies considered below, the energy gap is comparable to the corresponding squared natural distance; hence an energy bound of order $m^{-\alpha}$ gives a V-distance bound of order $m^{-\alpha/2}$ and, for $p \geq 2$, a $W^{1,p}$-error bound of order $m^{-\alpha/p}$. This conversion has the same form for every p, although the energy exponent may depend on p. We prove the equivalence for the standard pure, regularized, and reaction p-energies and apply the corresponding conversion to each of them.

Under the atomic source condition of Assumption 2.2, the directional smoothness estimate of Assumption 4.1, and a uniform positive selection strength, the ideal continuous CGA has an $O(m^{-1})$ energy bound. For finite-pool CGA, the one-step inequality quantifies separately the effects of continuous-dictionary coverage, score-evaluation error, radial active-space stationarity error, and coefficient-optimization error.

Local Hessian transfer combines Hessian coercivity, dictionary norm comparison, and a relative score condition to obtain a weak-OGA estimate in a frozen Hilbert metric.

The computed nested spaces also admit a finite-trajectory projection analysis. For a fixed finite symmetric dictionary and starting index m_0, an approximate source

representation with coefficient budget B_{entry} and remainder bound σ gives

$$r_{m_0+j}^2 \le \sigma^2 + \frac{4B_{\mathrm{entry}}^2}{W_j}, \qquad W_j = \sum_{m=m_0}^{m_0+j-1} \frac{\underline{\theta}_m^2}{\ell_m^2}, \qquad W_j > 0.$$

Here r_m is the projection error, ℓ_m is the innovation norm, and $0 \le \underline{\theta}_m \le \theta_m$ measures the selected score relative to the maximum frozen-dictionary score S_m^{fin}. For the one-dimensional $p = 4$ energies, an exact quartic expansion converts projection estimates into energy bounds. The analysis also gives an initial-value form of the source bound. Under the power-type condition $\gamma_m^2 \ge c_0 r_m^{2+2/\alpha}$, with $\gamma_m = \theta_m S_m^{\mathrm{fin}}/\ell_m$, the projection recurrence yields

$$r_{m_0+j}^2 \le \left(r_{m_0}^{-2/\alpha} + c_0 j/\alpha \right)^{-\alpha}.$$

The quartic comparison then supplies the corresponding energy and metric bounds.

The numerical study covers eight cases: linear reaction–diffusion, cubic and hyperbolic-sine semilinear energies, and pure, regularized, and reaction p-energies. Five cases compare finite-pool CGA, random-feature models (RFM), and finite elements (FEM) at prescribed coefficient counts using a common evaluator. Random features follow the fixed-feature construction of [12]; conditioning of linearized neural trial spaces is discussed by [4, 11]. At the five reported endpoints, CGA has smaller Sobolev errors than the metric-valid RFM medians and smaller errors than the P3 FEM references in four cases.

The finite-trajectory study uses the one-dimensional pure and regularized $p = 4$ models. Source quantities and comparison constants are calibrated on $N = 8, \dots, 32$ and held fixed during continuation to $N = 64$. The approximate source is constructed at the starting index from the candidate and auxiliary reference dictionaries. The computations evaluate the source bound, the quartic energy identity, and the power-type projection bound with the prescribed benchmark $\alpha_{\mathrm{bench}} = 6$.

Section 2 introduces the variational setting, dictionaries, and algorithms. Section 3 develops the energy families and natural-distance estimates. Section 4 presents finite-pool descent, local Hessian transfer, and finite-trajectory projection bounds. Section 5 reports the CGA experiments, and Section 6 compares the methods at prescribed coefficient counts. Section 7 discusses the main findings.

2. Variational setting, dictionaries, and algorithms. Let X be a real reflexive Banach space, and let $\mathcal{E} : X \to (-\infty, +\infty]$ be a proper convex functional with a unique minimizer u^* in the admissible space. Its effective domain is

$$\operatorname{dom} \mathcal{E} = \{v \in X : \mathcal{E}(v) < +\infty\}.$$

The descent arguments are carried out on an open neighborhood of the relevant iterates where $\mathcal{E}$ is finite and continuously Fréchet differentiable. The energy space and topology of this neighborhood are specified in the corresponding result. For energies with a restricted effective domain, any additional topology is stated explicitly. Arguments involving a Hessian further require twice Fréchet differentiability on the stated neighborhood.

We denote the duality pairing between X^* and X by $\langle r, v \rangle$ and define the energy defect by

$$\delta(u) = \mathcal{E}(u) - \mathcal{E}(u^*), \qquad \delta_m = \delta(G_m).$$

2.1. Ridge dictionaries and admissibility. For an integer activation order $k \geq 1$, a raw ridge atom has the form

$$\varphi_{w,b}(x) = \mathrm{ReLU}(w \cdot x + b)^k, \qquad (w, b) \in \mathbb{R}^d \times \mathbb{R}.$$

An admissible dictionary consists of nonzero ridge functions satisfying the constraints of X, after the corresponding linear projection. We use a symmetric normalized dictionary $\mathcal{D}_X \subset X$:

$$g \in \mathcal{D}_X \Longrightarrow -g \in \mathcal{D}_X, \qquad \|g\|_X = 1 \quad (g \in \mathcal{D}_X).$$

When density is needed, we also require $\overline{\operatorname{span} \mathcal{D}_X}^X = X$. We abbreviate $\mathcal{D}_X$ to $\mathcal{D}$ unless a different normalization is specified.

For pure-gradient Neumann energies the admissible space is

$$X_{p,\circ} = \left\{ v \in W^{1,p}(\Omega) : \int_\Omega v \, \mathrm{d}x = 0 \right\},$$

where $\Omega \subset \mathbb{R}^d$ is the bounded, connected Lipschitz domain considered in Section 3. Each ridge is projected onto $X_{p,\circ}$ before normalization. On a subdictionary with projected norms bounded below by a positive constant, uniform bounds for the projected atoms transfer to the normalized dictionary. The analytical dictionary consists of the nonzero projected atoms.

DEFINITION 2.1 (Atomic hull and atomic norm). *For a symmetric normalized dictionary $\mathcal{D} \subset X$, define*

$$K_1(\mathcal{D}) = \overline{\left\{ \sum_{j=1}^{J} c_j g_j : J < \infty,\ g_j \in \mathcal{D},\ \sum_{j=1}^{J} |c_j| \leq 1 \right\}}^X,$$

and

$$\|v\|_{\mathcal{A}(\mathcal{D})} = \inf\{B > 0 : v/B \in K_1(\mathcal{D})\}.$$

For a bounded set $K \subset X$, its n-th dyadic entropy number is

$$\varepsilon_n(K; X) = \inf \left\{ h > 0 : \exists x_1, \ldots, x_{2^{n-1}} \in X, K \subset \bigcup_{j=1}^{2^{n-1}} (x_j + hB_X) \right\}, \qquad n \geq 1,$$

where $B_X = \{x \in X : \|x\|_X \leq 1\}$. Thus the covering balls may have centers in X. We write $A \lesssim B$ when $A \leq CB$ with a constant independent of the iteration index; a subscript on $\lesssim$ lists the permitted parameter dependence, and $A \simeq B$ denotes the two corresponding inequalities.

ASSUMPTION 2.2 (Atomic source). *There is a finite $B_{\mathrm{src}} > 0$, with $B_{\mathrm{src}} \geq \|u^*\|_{\mathcal{A}(\mathcal{D})}$, such that $u^*/B_{\mathrm{src}} \in K_1(\mathcal{D})$.*

The source condition uses the continuous dictionary normalized in X. In one space dimension, Proposition 2.3 gives a sufficient condition for finite atomic norm.

PROPOSITION 2.3 (A one-dimensional ReLU^k source class). *Let $I = (0, 1)$, $k \geq 1$, $1 \leq p < \infty$, and $X = W^{1,p}(I)$. Suppose that the continuous dictionary contains the normalized nonzero truncated powers $(x - t)_+^k / \|(\cdot - t)_+^k\|_X$, $0 \leq t < 1$, together with*

normalized fully active ridge functions spanning $\mathbb{P}_k$, the space of polynomials of degree at most k on I. If $v \in W^{k+1,1}(I)$, then v belongs to the atomic space and

$$\|v\|_{\mathcal{A}(\mathcal{D})} \le C_{k,p} \left(\sum_{j=0}^{k} \left| v^{(j)}(0) \right| + \left\| v^{(k+1)} \right\|_{L^1(I)} \right),$$

where $C_{k,p}$ depends on the exponent p, the interval, and the normalization of the raw atoms.

The proof in Supplement S4 uses the integral remainder formula. Multivariate source and entropy estimates follow from shallow-network variation-space and width results under their domain and norm hypotheses [15, 16, 10].

LEMMA 2.4 (Projection and renormalization). *Let $P : X \to X_0$ be bounded and linear. If a subdictionary $\mathcal{D}_0 \subset \mathcal{D}$ satisfies*

$$0 < c_0 \le \|Pg\|_{X_0} \le C_0 < \infty \qquad (g \in \mathcal{D}_0),$$

and $\widetilde{\mathcal{D}}_0 = \{Pg/\|Pg\|_{X_0} : g \in \mathcal{D}_0\}$, then

$$PK_1(\mathcal{D}_0) \subset C_0 K_1(\widetilde{\mathcal{D}}_0).$$

The same computation also shows that a bounded linear map increases every entropy radius by at most $\|P\|$, while the inverse normalization loses at most the factor c_0^{-1}:

$$K_1(\widetilde{\mathcal{D}}_0) \subset c_0^{-1} P K_1(\mathcal{D}_0).$$

Proof sketch. For each atomic combination, write $Pg = \|Pg\|_{X_0}(Pg/\|Pg\|_{X_0})$ and use the upper and lower bounds on $\|Pg\|_{X_0}$. Mapping a cover through P multiplies its radius by at most $\|P\|$, while renormalization costs at most c_0^{-1}. ∎

COROLLARY 2.5 (Transfer of source and entropy inputs). *In the setting of Lemma 2.4, suppose $u^* = Pu^\dagger$, $u^\dagger/B \in K_1(\mathcal{D}_0)$, and*

$$\varepsilon_n(K_1(\mathcal{D}_0); X) \le C_{\mathrm{ent}} n^{-\beta}.$$

Then

$$\frac{u^*}{BC_0} \in K_1(\widetilde{\mathcal{D}}_0), \qquad \varepsilon_n(K_1(\widetilde{\mathcal{D}}_0); X_0) \le c_0^{-1} \|P\| \, C_{\mathrm{ent}} n^{-\beta}.$$

Thus a ReLU source/convex-hull entropy theorem on an extended parameter domain passes to the projected, renormalized feasible dictionary whenever the nondegeneracy bounds in Lemma 2.4 hold.

Proof sketch. Apply the two inclusions in Lemma 2.4 to the atomic representation of $u^\dagger$ and to an entropy cover of $K_1(\mathcal{D}_0)$. The source radius gains the factor C_0, and the covering radius gains $c_0^{-1}\|P\|$. ∎

2.2. Dictionary residuals. For $r \in X^*$, introduce the dictionary dual seminorm

$$\|r\|_{\mathcal{D}^*} = \sup_{g \in \mathcal{D}_X} |\langle r, g \rangle| .$$

This quantity measures the derivative in the available dictionary directions. It is a seminorm; if the dictionary has dense linear span, its vanishing implies that the derivative is zero.

2.3. Ideal and finite-pool Chebyshev greedy algorithms. When the derivative-score supremum is attained, the continuous CGA selects a maximizing atom and minimizes the energy over the enlarged trial space. Requiring only a prescribed fraction $t_m \in (0,1]$ of the supremum gives the weak CGA (wCGA), described in Algorithm 2.1.

Algorithm 2.1 Continuous weak Chebyshev greedy algorithm

Assume $\mathcal{E}(0) < \infty$, choose weakness numbers $t_m \in (0,1]$, and set $G_0 = 0$, $\Phi_0 = \{0\}$. Given G_m and $\Phi_m = \operatorname{span}\{g_1, \dots, g_m\}$:

(i) if $\|\mathcal{E}'(G_m)\|_{\mathcal{D}^*} = 0$, stop; otherwise choose $g_{m+1} \in \mathcal{D}_X$ with descent sign such that

$$-\langle \mathcal{E}'(G_m), g_{m+1} \rangle \geq t_m \|\mathcal{E}'(G_m)\|_{\mathcal{D}^*};$$

(ii) set $\Phi_{m+1} = \operatorname{span}(\Phi_m, g_{m+1})$ and compute

$$G_{m+1} = \operatorname*{arg\,min}_{v \in \Phi_{m+1}} \mathcal{E}(v).$$

The choice $t_m = 1$ corresponds to exact selection, available, for example, on a compact nondegenerate parameter set with continuous score. Since changing the sign of an atom leaves its linear span unchanged, the selected atom can be assigned its descent sign.

Exact reoptimization yields the variational orthogonality

$$\langle \mathcal{E}'(G_m), v \rangle = 0 \qquad (v \in \Phi_m). \tag{2.1}$$

The finite-pool algorithm uses a finite admissible pool $\mathcal{P}_m$ and, at the current computed state $\widehat{G}_m$, a numerically evaluated score $\widetilde{s}_m : \mathcal{P}_m \to [0, \infty)$ approximating $g \mapsto |\left\langle \mathcal{E}'(\widehat{G}_m), g \right\rangle|$. Coefficient minimization is performed to a finite tolerance.

Algorithm 2.2 Finite-pool inexact CGA

Set $\widehat{G}_0 = 0$, $\widehat{\Phi}_0 = \{0\}$. At step m, form a finite admissible pool $\mathcal{P}_m \subset \mathcal{D}$. Let

$$s_m(g) = \left| \left\langle \mathcal{E}'(\widehat{G}_m), g \right\rangle \right|, \qquad \widetilde{s}_m(g) \geq 0.$$

Choose $t_m^{\mathrm{pool}} \in (0,1]$ and $\widehat{g}_{m+1} \in \mathcal{P}_m$ satisfying

$$\widetilde{s}_m(\widehat{g}_{m+1}) \geq t_m^{\mathrm{pool}} \max_{g \in \mathcal{P}_m} \widetilde{s}_m(g),$$

give it the true descent sign, and define $\widehat{\Phi}_{m+1} = \operatorname{span}(\widehat{\Phi}_m, \widehat{g}_{m+1})$. Compute a coefficient vector whose state obeys

$$\mathcal{E}(\widehat{G}_{m+1}) \leq \inf_{v \in \widehat{\Phi}_{m+1}} \mathcal{E}(v) + \varepsilon_m^{\mathrm{opt}}, \qquad \varepsilon_m^{\mathrm{opt}} \geq 0.$$

For each run, the implementation records the reason for termination.

2.4. Finite-pool error quantities. The score error and continuous-dictionary coverage are quantified by

$$\sup_{g\in\mathcal{P}_m} |\widetilde{s}_m(g) - s_m(g)| \le \xi_m, \tag{2.2}$$

$$\tau_m = \frac{\max_{g\in\mathcal{P}_m} s_m(g)}{\sup_{g\in\mathcal{D}} s_m(g)}, \qquad 0 \le \tau_m \le 1, \tag{2.3}$$

with $\tau_m = 1$ when the denominator is zero. The convergence estimates specify a positive lower bound on the effective selection strength $t_m^{\mathrm{pool}}\tau_m$.

LEMMA 2.6 (True score selected from an evaluated pool). *Under* (2.2) *and* (2.3),

$$s_m(\widehat{g}_{m+1}) \ge t_m^{\mathrm{pool}}\tau_m \left\|\mathcal{E}'(\widehat{G}_m)\right\|_{\mathcal{D}^*} - (1 + t_m^{\mathrm{pool}})\xi_m. \tag{2.4}$$

Proof sketch. Apply the score-error bound before and after the approximate maximization, and express the finite-pool maximum through τ_m and the continuous score supremum. ∎

For coefficient optimization with the discrete energy $\mathcal{E}_h$, the following estimate relates the correction tolerance to the discrepancy between $\mathcal{E}_h$ and $\mathcal{E}$ on a suitable sublevel set of the current trial space $\widehat{\Phi}_{m+1}$.

Let $G_{m+1}^\sharp$ denote an exact minimizer of $\mathcal{E}$ over $\widehat{\Phi}_{m+1}$. For an energy level Λ_m^E chosen so that both $G_{m+1}^\sharp$ and the computed state $\widehat{G}_{m+1}$ are included, define the relevant sublevel set and its active-space restriction by

$$\mathcal{L}_m = \{v \in X : \mathcal{E}(v) \le \Lambda_m^E\}, \qquad \mathcal{L}_m^{\mathrm{act}} = \widehat{\Phi}_{m+1} \cap \mathcal{L}_m. \tag{2.5}$$

Thus $\mathcal{L}_m^{\mathrm{act}}$ is the part of the current finite-dimensional trial space on which the discrete energy $\mathcal{E}_h$ and the variational energy $\mathcal{E}$ are compared; in particular, $\inf_{v\in\mathcal{L}_m^{\mathrm{act}}} \mathcal{E}(v) = \mathcal{E}(G_{m+1}^\sharp)$.

LEMMA 2.7 (Transfer from the discrete energy to the analysis energy). *Suppose on the active-space sublevel set* $\mathcal{L}_m^{\mathrm{act}}$ *in* (2.5) *that*

$$\sup_{v\in\mathcal{L}_m^{\mathrm{act}}} |\mathcal{E}_h(v) - \mathcal{E}(v)| \le \zeta_m^E.$$

If the computed state satisfies

$$\mathcal{E}_h(\widehat{G}_{m+1}) \le \inf_{v\in\mathcal{L}_m^{\mathrm{act}}} \mathcal{E}_h(v) + \varepsilon_m^{\mathrm{opt,h}},$$

then Algorithm 2.2 *holds with* $\varepsilon_m^{\mathrm{opt}} \le \varepsilon_m^{\mathrm{opt,h}} + 2\zeta_m^E$.

Proof sketch. Compare the computed state with the exact active minimizer in the common sublevel set. The uniform energy discrepancy costs at most ζ_m^E at each endpoint, so the two discrepancies add to $2\zeta_m^E$. ∎

3. Natural geometry for variational energies. Let $\Omega \subset \mathbb{R}^d$ be bounded, connected, and Lipschitz. We consider diffusion and reaction equations with natural Neumann boundary conditions. The linear and semilinear equations are

$$-\Delta u + u = f, \qquad -\Delta u + u^3 = f, \qquad -\Delta u + \sinh u = f.$$

The standard p-structure models are

$$-\operatorname{div} A_p(\nabla u) = f, \quad -\operatorname{div} A_{p,\varepsilon}(\nabla u) = f, \quad -\operatorname{div} A_p(\nabla u) + |u|^{p-2}u = f,$$

where $A_p(\xi) = |\xi|^{p-2}\xi$, $A_p(0) = 0$, and $A_{p,\varepsilon}(\xi) = (\varepsilon^2 + |\xi|^2)^{(p-2)/2}\xi$. Here $\varepsilon \geq 0$ is fixed, with $\varepsilon > 0$ for the regularized model. The boundary flux is zero. For the pure and regularized Neumann problems, the load annihilates constants and the zero-mean representative is used. The equations are understood weakly when the load is not a regular function.

3.1. Variational energies and feasible spaces. The three p-energies are defined in (3.4)–(3.6). The remaining models have the following energies, written for $d \leq 2$:

$$\mathcal{E}_{\mathrm{lin}}(u) = \int_\Omega \left(\frac{1}{2} |\nabla u|^2 + \frac{1}{2} u^2 \right) \mathrm{d}x - \langle f, u \rangle , \tag{3.1}$$

$$\mathcal{E}_{\mathrm{cub}}(u) = \int_\Omega \left(\frac{1}{2} |\nabla u|^2 + \frac{1}{4} u^4 \right) \mathrm{d}x - \langle f, u \rangle , \tag{3.2}$$

$$\mathcal{E}_{\sinh}(u) = \int_\Omega \left(\frac{1}{2} |\nabla u|^2 + \cosh u - 1 \right) \mathrm{d}x - \langle f, u \rangle . \tag{3.3}$$

Here $\langle f, u \rangle$ denotes the duality pairing between the relevant energy space and its dual. When the load admits an integral representation, $\langle f, u \rangle = \int_\Omega f u \, \mathrm{d}x$. Subtracting 1 from $\cosh u$ normalizes the potential to zero at $u = 0$ and leaves the first variation, hence the Euler–Lagrange equation, unchanged.

Assumption 3.1 (Data and feasible spaces for the six models). *The following model-specific conditions are imposed.*

(i) *For the linear and cubic models, $f \in (H^1(\Omega))^*$. The sinh energy has the same load space and is treated as an extended-real convex integral functional with nonempty effective domain. Its local derivative and Hessian estimates are formulated on an L^∞-bounded neighborhood.*

(ii) *For the pure and regularized p-energies, $f \in X_{p,\circ}^*$, or equivalently the Neumann load annihilates constants before restriction to $X_{p,\circ}$.*

(iii) *For the reaction p-energy, $f \in (W^{1,p})^*$. In the regularized model, the parameter ε in* (3.5) *is fixed and positive.*

The local Hessian structure of each model is needed for the transfer estimates of Subsection 4.2. The bilinear forms at the exact solution are

$$a_{\mathrm{lin}}(v, w) = \int_\Omega (\nabla v \cdot \nabla w + vw) \, \mathrm{d}x,$$

$$a_{\mathrm{cub},*}(v, w) = \int_\Omega (\nabla v \cdot \nabla w + 3(u^*)^2 vw) \, \mathrm{d}x,$$

$$a_{\sinh,*}(v, w) = \int_\Omega (\nabla v \cdot \nabla w + \cosh(u^*) vw) \, \mathrm{d}x.$$

Proposition 3.2 (Well-posedness). *Under Assumption* 3.1, (3.1)–(3.3) *have unique minimizers in $H^1(\Omega)$;* (3.4) *and* (3.5) *have unique minimizers in $X_{p,\circ}$; and* (3.6) *has a unique minimizer in $W^{1,p}(\Omega)$.*

Proof sketch. The quadratic, strictly convex, and p-growth terms give coercivity on the stated feasible spaces; strict convexity gives uniqueness. The pure Neumann nullspace is removed by the zero-mean restriction. ∎

For a p-diffusion density, the Hessian matrix in gradient variables is

$$|\xi|^{p-2} I + (p-2)\,|\xi|^{p-4}\,\xi \otimes \xi$$

at nonzero ξ in the pure case, with $|\xi|^2$ replaced by $\varepsilon^2 + |\xi|^2$ in the regularized case. The local Hessian statements that follow concern $p \geq 2$.

PROPOSITION 3.3 (Sufficient continuous local Hessian conditions). *In dimensions $d \leq 2$, the cubic Hessian is locally Lipschitz as a bilinear form on H^1 and is coercive at any solution $u^* \not\equiv 0$. The sinh Hessian is coercive and locally Lipschitz on an L^∞-bounded neighborhood, with state perturbations measured in H^1. For fixed $\varepsilon > 0$, the regularized p-Hessian is uniformly elliptic with lower bound proportional to ε^{p-2}, and locally Lipschitz on bounded subsets of $W^{1,\infty}$.*

Proof sketch. In dimensions $d \leq 2$, Sobolev embedding controls the cubic coefficient on bounded H^1 neighborhoods, while an L^∞ bound controls the sinh coefficient. Differentiating the regularized radial flux gives uniform ellipticity and bounded derivative on bounded-gradient sets. ∎

PROPOSITION 3.4 (Restricted stability for the regularized p-energy). *Let $Y_M \subset X_{p,\circ} \cap W^{1,\infty}(\Omega)$ be finite dimensional, equipped with a norm $\|\cdot\|_{0,M}$, and let $U_M \subset Y_M$ be a bounded convex set. For fixed $p \geq 2$ and $\varepsilon > 0$, there are constants $\mu_M > 0$ and $L_M < \infty$ such that, for $u, z \in U_M$ and $v, w \in Y_M$,*

$$\begin{aligned} \mathcal{E}''(u)[v,v] &\geq \mu_M \|v\|_{0,M}^2\,, \\ |(\mathcal{E}''(u) - \mathcal{E}''(z))[v,w]| &\leq L_M \|u-z\|_{0,M} \|v\|_{0,M} \|w\|_{0,M}\,. \end{aligned}$$

The constants depend on U_M, Y_M, the reference norm, p, and ε.

Proof sketch. Finite-dimensional norm equivalence gives a Hessian-Lipschitz constant on the bounded set. Uniform ellipticity and the zero-mean condition give coercivity in the reference norm. ∎

The linear, cubic, and sinh models are treated through their quadratic or local Hessian structure. For the regularized p-energy, uniform ellipticity gives the stated finite-dimensional estimates. The pure and reaction p-energies are described by the global energy–distance relations and the finite-trajectory projection analysis.

3.2. Standard p-structure and natural distance. For standard p-structure energies, the pointwise inequalities for the diffusion density connect the CGA energy error to its natural distance. Integrating these inequalities gives energy-to-metric estimates and the corresponding $W^{1,p}$ conversion for the three energy families below. Let $\Omega \subset \mathbb{R}^d$ be bounded, connected, and Lipschitz, and let $p \geq 2$. On the zero-mean space $X_{p,\circ}$, define the pure and regularized energies

$$\mathcal{E}_p(u) = \int_\Omega \frac{1}{p} |\nabla u|^p \,\mathrm{d}x - \langle f, u\rangle\,, \tag{3.4}$$

$$\mathcal{E}_{p,\varepsilon}(u) = \int_\Omega \frac{(\varepsilon^2 + |\nabla u|^2)^{p/2} - \varepsilon^p}{p} \,\mathrm{d}x - \langle f, u\rangle\,, \qquad \varepsilon > 0. \tag{3.5}$$

On $W^{1,p}(\Omega)$, define the reaction energy

$$\mathcal{E}_{p,R}(u) = \int_\Omega \left(\frac{1}{p} |\nabla u|^p + \frac{1}{p} |u|^p \right) \mathrm{d}x - \langle f, u\rangle\,. \tag{3.6}$$

For a differentiable convex functional F, its Bregman divergence is

$$D_F(u,v) = F(u) - F(v) - \langle F'(v), u - v\rangle \geq 0. \tag{3.7}$$

It measures the part of the energy increment not accounted for by the tangent functional at v. In general $D_F(u,v) \neq D_F(v,u)$, so it is not a metric. The same notation is used pointwise for a convex density: $D_\phi(a,b) = \phi(a) - \phi(b) - \nabla\phi(b)\cdot(a-b)$.

For scalar arguments set $V(s) = |s|^{(p-2)/2}\, s$, and for vector arguments set

$$V(\xi) = |\xi|^{(p-2)/2}\,\xi, \qquad V_\varepsilon(\xi) = (\varepsilon^2 + |\xi|^2)^{(p-2)/4}\xi.$$

The corresponding natural distances are

$$\begin{aligned} d_V(u,v)^2 &= \|V(\nabla u) - V(\nabla v)\|_{L^2}^2\,, \\ d_{V,\varepsilon}(u,v)^2 &= \|V_\varepsilon(\nabla u) - V_\varepsilon(\nabla v)\|_{L^2}^2\,, \\ d_{V,R}(u,v)^2 &= d_V(u,v)^2 + \|V(u) - V(v)\|_{L^2}^2\,. \end{aligned}$$

We use $\mathcal{E}_\bullet$ and $d_{V,\bullet}$ only as paired abbreviations for one of $(\mathcal{E}_p, d_V)$, $(\mathcal{E}_{p,\varepsilon}, d_{V,\varepsilon})$, or $(\mathcal{E}_{p,R}, d_{V,R})$. The scalar map applies to the reaction term and the vector map to gradients. The map V turns the weighted quadratic structure of a p-operator into an L^2-type distance [1, 6, 8].

Throughout this section, $A \lesssim_p B$ means $A \leq C(p)B$, and $A \simeq_p B$ means that both inequalities hold with positive constants depending only on p; a further subscript lists any additional permitted dependence.

LEMMA 3.5 (Pointwise p-structure). *For $p \geq 2$, constants depending only on p satisfy, for every $a, b \in \mathbb{R}^d$,*

$$\frac{|a|^p}{p} - \frac{|b|^p}{p} - |b|^{p-2}\, b\cdot(a-b) \simeq_p (|a| + |b|)^{p-2}\,|a-b|^2 \simeq_p |V(a) - V(b)|^2\,, \tag{3.8}$$

$$|a-b|^p \lesssim_p |V(a) - V(b)|^2\,. \tag{3.9}$$

The analogous statements hold for the regularized density and V_ε, with $\varepsilon > 0$ fixed.

Proof sketch. These are the standard monotonicity and radial-growth inequalities for $\xi \mapsto |\xi|^{p-2}\xi$; the regularized statements follow from the same calculation with $\varepsilon^2 + |\xi|^2$. ∎

THEOREM 3.6 (Energy–natural-distance equivalence). *For each of the three energy–space pairs*

$$(\mathcal{E}_p, X_{p,\circ}), \qquad (\mathcal{E}_{p,\varepsilon}, X_{p,\circ}), \qquad (\mathcal{E}_{p,R}, W^{1,p}(\Omega)),$$

let u^* be the minimizer. Then, for every admissible u, respectively,

$$\mathcal{E}_p(u) - \mathcal{E}_p(u^*) \simeq_p d_V(u,u^*)^2, \tag{3.10}$$

$$\mathcal{E}_{p,\varepsilon}(u) - \mathcal{E}_{p,\varepsilon}(u^*) \simeq_{p,\varepsilon} d_{V,\varepsilon}(u,u^*)^2, \tag{3.11}$$

$$\mathcal{E}_{p,R}(u) - \mathcal{E}_{p,R}(u^*) \simeq_p d_{V,R}(u,u^*)^2. \tag{3.12}$$

In the corresponding spaces,

$$\|u - u^*\|_{W^{1,p}}^p \lesssim_{p,\Omega,\varepsilon} d_{V,\bullet}(u,u^*)^2, \tag{3.13}$$

where the constants in the pure and reaction cases are independent of ε.

Proof sketch. Integrate the pointwise Bregman equivalences for the gradient and reaction terms. The weak optimality of u^* cancels the load term, and Poincaré's inequality controls the zero-mean representative in the pure and regularized cases. ∎

COROLLARY 3.7 (Metric exponents). *Fix any one of the three energy–space–distance triples in Theorem* 3.6. *If a sequence G_m in that feasible space satisfies*

$$\mathcal{E}_\bullet(G_m) - \mathcal{E}_\bullet(u^*) = O(m^{-\alpha}) \tag{3.14}$$

for some $\alpha > 0$, then

$$d_{V,\bullet}(G_m, u^*) = O(m^{-\alpha/2}), \qquad \|G_m - u^*\|_{W^{1,p}} = O(m^{-\alpha/p}). \tag{3.15}$$

Proof sketch. Apply the two-sided energy–distance equivalence and the $W^{1,p}$ control in Theorem 3.6 to the assumed energy bound, then take the corresponding square or p-th root. ∎

Remark 3.8 (Model-specific geometry). The pure, regularized, and reaction p-energies satisfy the Bregman–V equivalence in Lemma 3.5. The linear and semilinear models are described by their quadratic and local Hessian geometries.

4. Convergence analysis and finite-trajectory projection bounds. We first derive finite-pool descent estimates in terms of pool coverage, score evaluation, radial stationarity, and coefficient correction. Local Hessian transfer then connects nonlinear selection with approximation in a frozen Hilbert metric. Finally, source representations and normalized innovations give projection bounds for nested spaces, which the quartic energy identity converts into nonlinear energy estimates.

Throughout, m counts accepted independent directions, m_0 is a starting index, $\widehat{G}_m$ is a computed state, $G_m^\sharp$ an exact active-space minimizer, and z_m a frozen orthogonal projection. All derivatives act on the specified feasible linear space.

4.1. Finite-pool descent and the atomic energy rate. Let X be a real Banach space and $\mathcal{E} : X \to (-\infty, +\infty]$ a proper convex energy with minimizer u^*. Assume that $\mathcal{E}$ is finite and continuously differentiable on an open neighborhood containing the states and dictionary segments used below. For a symmetric unit dictionary $\mathcal{D}_X$, set

$$\widehat{\delta}_m = \mathcal{E}(\widehat{G}_m) - \mathcal{E}(u^*), \qquad S_m^X = \sup_{g \in \mathcal{D}_X} |\left\langle \mathcal{E}'(\widehat{G}_m), g \right\rangle|.$$

For a finite pool $\mathcal{P}_m \subset \mathcal{D}_X$ and $0 < t_m^{\mathrm{pool}} \le 1$, write

$$|\widetilde{s}_m(g) - s_m(g)| \le \xi_m, \qquad \widetilde{s}_m(g_{m+1}) \ge t_m^{\mathrm{pool}} \max_{g \in \mathcal{P}_m} \widetilde{s}_m(g),$$

where the score s_m and the coverage factor τ_m are defined in Algorithm 2.2 and (2.3), respectively.

ASSUMPTION 4.1 (Directional quadratic upper bound). *For some $L > 0$, every m, every $g \in \mathcal{D}_X$, and $|\lambda| \le 1$,*

$$\mathcal{E}(\widehat{G}_m + \lambda g) \le \mathcal{E}(\widehat{G}_m) + \lambda \left\langle \mathcal{E}'(\widehat{G}_m), g \right\rangle + \tfrac{L}{2}\lambda^2, \tag{4.1}$$

Assume that the expanded active space contains the signed step used below and that the inexact active correction satisfies

$$\mathcal{E}(\widehat{G}_{m+1}) \le \inf_{v\in\widehat{\Phi}_{m+1}} \mathcal{E}(v) + \varepsilon_m^{\mathrm{opt}}, \qquad \varepsilon_m^{\mathrm{opt}} \ge 0.$$

The radial active-space stationarity defect is denoted by

$$\left|\left\langle \mathcal{E}'(\widehat{G}_m), \widehat{G}_m \right\rangle\right| \le \eta_m^{\mathrm{stat}}. \tag{4.2}$$

This is the radial component of the active-space stationarity residual; the full active-space dual KKT residual is a separate quantity.

Proposition 4.2 (Finite-pool descent). *Under these conditions,*

$$\widehat{\delta}_{m+1} \le \widehat{\delta}_m - H_L(a_m) + \varepsilon_m^{\mathrm{opt}}, \qquad a_m = \left[t_m^{\mathrm{pool}}\tau_m S_m^X - (1+t_m^{\mathrm{pool}})\xi_m\right]_+, \tag{4.3}$$

where

$$H_L(a) = \max_{0\le\lambda\le 1}\left(\lambda a - \frac{L}{2}\lambda^2\right), \qquad a \ge 0.$$

Proof. The score error is inserted before and after the finite-pool maximization. Testing the enlarged active space with a signed step of length $\min\{1, a_m/L\}$ gives the displayed decrease, and the correction error is added once. ∎

If $B_{\mathrm{src}} > 0$, u^*/B_{src} belongs to the closed absolutely convex hull of $\mathcal{D}_X$, and the radial defect satisfies (4.2), convexity gives

$$\widehat{\delta}_m \le \eta_m^{\mathrm{stat}} + B_{\mathrm{src}} S_m^X. \tag{4.4}$$

This relation bounds the energy defect by the dictionary score and the radial stationarity residual.

Lemma 4.3 (Nonautonomous scalar recurrence). *If $x_m \ge 0$ and $x_{m+1} \le x_m - b_m x_m^q$ with $q > 1$ and $b_m \ge 0$, then, for $x_0 > 0$,*

$$x_n \le \left[x_0^{1-q} + (q-1)\sum_{m=0}^{n-1} b_m\right]^{-1/(q-1)}. \tag{4.5}$$

If $x_0 = 0$, the sequence vanishes.

Proof. Apply convexity to $x \mapsto x^{1-q}$ while the sequence is positive and sum the resulting increments; once a term vanishes, nonnegativity gives the remaining cases. ∎

Corollary 4.4 (Atomic energy rate). *Assume the hypotheses of Proposition 4.2 and $u^* \in B_{\mathrm{src}} K_1(\mathcal{D}_X)$ with $B_{\mathrm{src}} > 0$. Suppose that, for some m_0, $\underline{t} > 0$, and $0 \le \vartheta, \zeta < 1$, the radial defect satisfies (4.2) and*

$$t_m^{\mathrm{pool}}\tau_m \ge \underline{t}, \qquad \eta_m^{\mathrm{stat}} + \frac{B_{\mathrm{src}}(1+t_m^{\mathrm{pool}})}{t_m^{\mathrm{pool}}\tau_m}\xi_m \le \vartheta\widehat{\delta}_m, \qquad \varepsilon_m^{\mathrm{opt}} \le \zeta H_L(a_m)$$

for all $m \ge m_0$. If $\widehat{\delta}_{m_0} > 0$, set

$$b = \frac{\underline{t}(1-\vartheta)}{B_{\mathrm{src}}}, \qquad A = b\widehat{\delta}_{m_0}, \qquad c = \frac{(1-\zeta)b^2}{2\max\{L, A\}}.$$

Then

$$\widehat{\delta}_{m_0+j} \le \left(\widehat{\delta}_{m_0}^{-1} + cj\right)^{-1}, \qquad j \ge 0,$$

and hence $\widehat{\delta}_m = O((m-m_0+1)^{-1})$. If $\widehat{\delta}_{m_0} = 0$, then $\widehat{\delta}_m = 0$ for every $m \ge m_0$.

Proof. Proposition 4.2 and the correction-error bound give

$$\widehat{\delta}_{m+1} \le \widehat{\delta}_m - (1-\zeta)H_L(a_m),$$

so the energy errors are nonincreasing. Equation (4.4) and the relative-error condition imply $a_m \ge b\widehat{\delta}_m$. Since $0 \le b\widehat{\delta}_m \le A$ and H_L is nondecreasing,

$$H_L(a_m) \ge H_L(b\widehat{\delta}_m) \ge \frac{b^2\widehat{\delta}_m^2}{2\max\{L,A\}}.$$

Thus $\widehat{\delta}_{m+1} \le \widehat{\delta}_m - c\widehat{\delta}_m^2$. Apply Lemma 4.3 to $x_j = \widehat{\delta}_{m_0+j}$ with $q=2$. The zero initial-error case follows from monotonicity. ∎

The ideal continuous method is recovered by setting $\tau_m = 1$ and $\xi_m = \eta_m^{\mathrm{stat}} = \varepsilon_m^{\mathrm{opt}} = 0$, with selection strength bounded below.

For the standard pure, regularized, and reaction p-energies, Theorem 3.6 and Corollary 3.7 give energy–distance equivalence and the $W^{1,p}$ conversion. Hence an energy estimate $\widehat{\delta}_m = O(m^{-\alpha})$ implies

$$d_{V,\bullet}(\widehat{G}_m, u^*) = O(m^{-\alpha/2}), \qquad \|\widehat{G}_m - u^*\|_{W^{1,p}} = O(m^{-\alpha/p}).$$

4.2. Local Hessian transfer. The local transfer result connects nonlinear energy minimization with Hilbert-space approximation. Let H be a Hilbert space on which $\mathcal{E}$ is twice Fréchet differentiable in an open convex neighborhood of its minimizer u^*. Let $A_* = \mathcal{E}''(u^*)$ be bounded, symmetric, and coercive, and set $a_*(v,w) = \langle A_* v, w\rangle$, $\|v\|_*^2 = a_*(v,v)$. Assume the closed ball $\|u-u^*\|_* \le R_*$ lies in this neighborhood and

$$|\langle[\mathcal{E}''(u) - A_*]v, w\rangle| \le L_*\|u-u^*\|_*\|v\|_*\|w\|_*, \qquad L_*R_* \le \tfrac{1}{2}. \tag{4.6}$$

For a finite-dimensional subspace $\Phi \subset H$, let $P_\Phi : H \to \Phi$ be the a_*-orthogonal projection, characterized by

$$a_*(u - P_\Phi u, v) = 0 \qquad (u \in H,\ v \in \Phi).$$

Set $z_\Phi = P_\Phi u^*$, and let $G_\Phi^\sharp$ be the exact minimizer of $\mathcal{E}$ over Φ. Assume that both z_Φ and $G_\Phi^\sharp$ lie in this ball.

LEMMA 4.5 (Taylor and projection estimates). *Under* (4.6), *for u in this ball,*

$$\left|\mathcal{E}(u) - \mathcal{E}(u^*) - \tfrac{1}{2}\|u-u^*\|_*^2\right| \le \tfrac{L_*}{6}\|u-u^*\|_*^3, \tag{4.7}$$

$$\|\mathcal{E}'(u) - A_*(u-u^*)\|_{*'} \le \tfrac{L_*}{2}\|u-u^*\|_*^2, \tag{4.8}$$

$$\|G_\Phi^\sharp - z_\Phi\|_* \le L_*\|z_\Phi - u^*\|_*^2. \tag{4.9}$$

Proof. Taylor's formula gives the first two inequalities. The third follows by testing the active-space first-order condition with $G_\Phi^\sharp - z_\Phi$ and absorbing the term $L_*R_*\|G_\Phi^\sharp - z_\Phi\|_*^2$. ∎

Assume the algorithmic dictionary $\mathcal{D}_X \subset H$ satisfies

$$0 < c_* \le \|g\|_* \le C_* < \infty \quad (g \in \mathcal{D}_X), \qquad \mathcal{D}_* = \{g/\|g\|_* : g \in \mathcal{D}_X\}. \tag{4.10}$$

Let $\Phi_{m+1} = \Phi_m + \operatorname{span}\{g_{m+1}\}$, where $g_{m+1} \in \mathcal{D}_X \setminus \Phi_m$, and set $P_m = P_{\Phi_m}$. Set $z_m = P_m u^*$ and define $S_m^* = \sup_{d\in\mathcal{D}_*} |\langle A_*(z_m - u^*), d\rangle|$. At the exact active-space minimizer $G_m^\sharp$ define the score defect

$$\eta_m^H = \sup_{g\in\mathcal{D}_X} \left|\langle \mathcal{E}'(G_m^\sharp), g\rangle - a_*(z_m - u^*, g)\right|.$$

Lemma 4.5 yields $\eta_m^H \le C_T \|z_m - u^*\|_*^2$ on the stated ball, with C_T uniform over the specified dictionary.

PROPOSITION 4.6 (Relative score transfer). *If nonlinear selection at $G_m^\sharp$ over $\mathcal{D}_X$ has strength $t \in (0,1]$ and $0 < t_0 < tc_*/C_*$, then*

$$(1+t)\eta_m^H \le (tc_* - t_0 C_*) S_m^* \tag{4.11}$$

implies that the selected direction satisfies

$$|a_*(z_m - u^*, g_{m+1}/\|g_{m+1}\|_*)| \ge t_0 S_m^*.$$

Proof. The selected unnormalized frozen score is bounded below by $tc_* S_m^* - (1+t)\eta_m^H$; division by the selected frozen norm costs at most C_*. This proves the claim. □

To make the entropy-to-rate step explicit, fix a restart index m_0 and set $Q_0 = I - P_{m_0}$, $H_0 = Q_0 H$, $f_0 = Q_0 u^*$, and $\mathcal{D}_{\mathrm{rst}} = Q_0 \mathcal{D}_* \setminus \{0\}$. The projected atoms retain their projected norms. Suppose $K = K_1(\mathcal{D}_{\mathrm{rst}})$ is compact and $f_0 \in B_* K$, $B_* > 0$. For weak OGA of strength t_0 in H_0, let $d_i \in \mathcal{D}_{\mathrm{rst}}$ be the selected atoms and Π_j the orthogonal projection onto the span of the first j selected atoms, with $\Pi_0 = 0$. The bounded-atom entropy argument of [9, Theorem 4.2] gives

$$\|f_0 - \Pi_j f_0\|_* \le \frac{B_*}{t_0} \frac{(j! V_j)^{1/j}}{\sqrt{j}} \varepsilon_{j+1}(K; H_0), \qquad V_j = \frac{\pi^{j/2}}{\Gamma(j/2+1)}. \tag{4.12}$$

Indeed, writing $y_i = \|f_0 - \Pi_i f_0\|_*^2$ and $\ell_i^{\mathrm{rst}} = \|(I - \Pi_{i-1}) d_i\|_*$, the source and weak selection give

$$y_i \le y_{i-1} - \frac{t_0^2 y_{i-1}^2}{B_*^2 (\ell_i^{\mathrm{rst}})^2}, \qquad y_j \le \frac{B_*^2}{t_0^2 j} \left(\prod_{i=1}^{j} \ell_i^{\mathrm{rst}} \right)^{2/j}.$$

The second bound follows by reciprocation and the arithmetic–geometric mean inequality while the errors are positive. The cross-polytope generated by $\{\pm d_1, \dots, \pm d_j\}$ lies in $\Pi_j K$ and has volume $(2^j/j!) \prod_i \ell_i^{\mathrm{rst}}$. A cover of K by 2^j balls yields $\prod_i \ell_i^{\mathrm{rst}} \le j! V_j \varepsilon_{j+1}(K; H_0)^j$, by taking the infimum over covering radii. This proves (4.12); zero errors persist. Stirling's formula bounds $(j! V_j)^{1/j}/\sqrt{j}$ uniformly.

THEOREM 4.7 (Hessian transfer after restart). *Assume (4.6), (4.10), and exact active-space minimization. At each step $m_0 \le m < m_0 + J$, suppose nonlinear selection has strength at least t and (4.11) holds with the same $t_0 > 0$. Assume the exact minimizers and frozen projections remain in the Hessian ball for $m_0 \le m \le m_0 + J$, K is compact, and*

$$f_0 \in B_* K_1(\mathcal{D}_{\mathrm{rst}}), \qquad \varepsilon_j(K_1(\mathcal{D}_{\mathrm{rst}}); H_0) \le C_{\mathrm{ent}} j^{-\beta}, \quad 1 \le j \le J+1,$$

with $B_*, C_{\mathrm{ent}} > 0$, $\beta > 0$. Then, for $1 \le j \le J$,

$$\|z_{m_0+j} - u^*\|_* \le C B_* C_{\mathrm{ent}} t_0^{-1} j^{-\beta}, \tag{4.13}$$

$$\mathcal{E}(G_{m_0+j}^\sharp) - \mathcal{E}(u^*) \le C_{\mathrm{nl}} j^{-2\beta}. \tag{4.14}$$

Here C is absolute and C_{nl} depends on $L_*, R_*, B_*, C_{\mathrm{ent}}, t_0$.

Proof. The relative score proposition gives weak frozen selection after projection by Q_0, so the restarted spaces form weak OGA on H_0. Apply (4.12). The projection lemma gives $\|G_{m_0+j}^\sharp - u^*\|_* \le (1 + L_* R_*) \|z_{m_0+j} - u^*\|_*$; the Taylor estimate then yields the energy bound. □

4.3. Approximate-source and quartic consequences. For the finite-trajectory analysis, let Y be a finite-dimensional real Hilbert space containing u^* and the active spaces in the prescribed range. Let $a_*(\cdot,\cdot)$ be its frozen inner product and P_m the associated orthogonal projection onto $\widehat{\Phi}_m$. Assume $\widehat{\Phi}_{m+1} = \widehat{\Phi}_m + \operatorname{span}\{d_{m+1}\}$ with $d_{m+1} \notin \widehat{\Phi}_m$, and set

$$z_m = P_m u^*, \qquad e_m = u^* - z_m, \qquad r_m = \|e_m\|_*.$$

Let $\mathcal{D}_{\mathrm{fin}} \subset Y$ be a finite symmetric dictionary containing every accepted direction in this range. For the selected direction d_{m+1} define its innovation norm $\ell_m = \|(I - P_m)d_{m+1}\|_* > 0$ and $S_m^{\mathrm{fin}} = \max_{d\in\mathcal{D}_{\mathrm{fin}}} |a_*(e_m, d)|$. Set $\theta_m = |a_*(e_m, d_{m+1})|/S_m^{\mathrm{fin}}$ when $S_m^{\mathrm{fin}} > 0$, and $\theta_m = 0$ otherwise. Assume that the restarted residual has the explicit approximate atomic representation

$$Q_0 u^* = \sum_{d\in\mathcal{D}_{\mathrm{fin}}} a_d Q_0 d + \rho_0, \qquad \sum_d |a_d| \le B_{\mathrm{entry}}, \qquad \|\rho_0\|_* \le \sigma, \quad Q_0 = I - P_{m_0}.$$

For any $0 \le \underline{\theta}_m \le \theta_m$, define $W_j = \sum_{m=m_0}^{m_0+j-1} \underline{\theta}_m^2/\ell_m^2$.

THEOREM 4.8 (Weighted finite-trajectory source bound). *For $x_0 = (r_{m_0}^2 - \sigma^2)_+ > 0$ and $B_{\mathrm{entry}} > 0$,*

$$(r_{m_0+j}^2 - \sigma^2)_+ \le \left(x_0^{-1} + \frac{W_j}{4B_{\mathrm{entry}}^2}\right)^{-1}, \qquad 0 \le j \le J. \tag{4.15}$$

If $x_0 = 0$ or $B_{\mathrm{entry}} = 0$, monotonicity gives $r_m \le \sigma$ on the range; for $W_j = 0$ the estimate is interpreted through projection monotonicity.

Proof. Pythagoras gives $r_m^2 - r_{m+1}^2 = |a_*(e_m, d_{m+1})|^2/\ell_m^2$, and the source representation gives $S_m^{\mathrm{fin}} \ge (r_m^2 - \sigma^2)_+/(2B_{\mathrm{entry}})$. When both $r_m > \sigma$ and $r_{m+1} > \sigma$, set $x_m = r_m^2 - \sigma^2$. Then

$$x_{m+1} \le x_m - \frac{\underline{\theta}_m^2 x_m^2}{4B_{\mathrm{entry}}^2 \ell_m^2}.$$

Taking reciprocals and summing gives the bound up to the first crossing of σ. At and after that crossing, projection monotonicity gives $(r_m^2 - \sigma^2)_+ = 0$. ∎

As a concrete example of converting projection bounds into nonlinear energy estimates, consider the one-dimensional pure and regularized quartic models. Let

$$X = \{u \in W^{1,4}(0,1) : \textstyle\int_0^1 u = 0\}, \qquad \mathcal{E}_\varepsilon(u) = \int_0^1 \left(\tfrac{\varepsilon^2}{2}|u'|^2 + \tfrac{1}{4}|u'|^4\right) \mathrm{d}x - \ell(u),$$

and assume $u^* \in X$ solves its weak equation. Let $Y \subset X$ contain these finite-dimensional spaces and define

$$a_*(v, w) = \int_0^1 (\varepsilon^2 + 3|(u^*)'|^2) v' w' \,\mathrm{d}x.$$

For $\varepsilon > 0$ this is positive definite on Y; for $\varepsilon = 0$ assume it is positive definite on all of Y. In this paragraph $\widehat{\delta}_m = \mathcal{E}_\varepsilon(\widehat{G}_m) - \mathcal{E}_\varepsilon(u^*)$. Then

$$\widehat{\delta}_m = \tfrac{1}{2}(r_m^2 + D_m^2) + R_m, \qquad R_m = \int_0^1 \left((u^*)' v_m^3 + \tfrac{1}{4} v_m^4\right) \mathrm{d}x, \tag{4.16}$$

where $D_m = \|\widehat{G}_m - z_m\|_*$ and $v_m = \widehat{G}_m' - (u^*)'$. The identity yields the following two-sided energy comparison.

PROPOSITION 4.9 (Quartic energy comparison). *If $q_{\mathrm{corr}} \geq 0$, $\rho \geq 0$, $D_m \leq q_{\mathrm{corr}} r_m$, and $|R_m| \leq \rho(r_m^2 + D_m^2)$ with $\rho < 1/2$, then*

$$(\tfrac{1}{2} - \rho) r_m^2 \leq \widehat{\delta}_m \leq (\tfrac{1}{2} + \rho)(1 + q_{\mathrm{corr}}^2) r_m^2. \tag{4.17}$$

Proof. Expand the quartic density about $(u^*)'$, cancel the linear term with the weak equation, and use orthogonality of the frozen projection to obtain (4.16); the comparison follows immediately. ∎

THEOREM 4.10 (Power-type finite-range bound). *Define the normalized frozen innovation decrease*

$$\gamma_m = \frac{|a_*(e_m, d_{m+1})|}{\ell_m}, \qquad r_{m+1}^2 = r_m^2 - \gamma_m^2.$$

Let $\alpha > 0$, $c_0 > 0$, and suppose $r_{m_0} > 0$. If $\gamma_m^2 \geq c_0 r_m^{2+2/\alpha}$ on $m_0 \leq m < m_0 + J$, then

$$r_{m_0+j}^2 \leq \left(r_{m_0}^{-2/\alpha} + \frac{c_0 j}{\alpha} \right)^{-\alpha}, \qquad 0 \leq j \leq J. \tag{4.18}$$

Proof. For positive successive residuals, apply convexity to $x^{-1/\alpha}$, with $x_m = r_m^2$, and sum. Once a residual is zero, nestedness keeps the subsequent residuals zero. ∎

COROLLARY 4.11 (Energy and metric bounds for quartic models). *Under the assumptions of Theorem 4.10, consider the one-dimensional pure or regularized quartic energy defined above. Suppose that the hypotheses of Proposition 4.9 hold for every $m_0 \leq m \leq m_0 + J$, with common constants $q_{\mathrm{corr}} \geq 0$ and $0 \leq \rho < 1/2$. Let U_j denote the right-hand side of (4.18), and set*

$$C_E = (1/2 + \rho)(1 + q_{\mathrm{corr}}^2).$$

Then, for $0 \leq j \leq J$,

$$\widehat{\delta}_{m_0+j} \leq C_E U_j, \qquad d_{V,\bullet}(\widehat{G}_{m_0+j}, u^*) \leq C_V (C_E U_j)^{1/2},$$

$$\|\widehat{G}_{m_0+j} - u^*\|_{W^{1,4}} \leq C_S (C_E U_j)^{1/4}.$$

Here $d_{V,\bullet}$ is the natural distance for the chosen model, and $C_V, C_S > 0$ are independent of j on the stated range.

Proof. Combine (4.18) with (4.17), and apply Theorem 3.6. ∎

5. CGA numerical experiments and finite-range estimates.

5.1. Model problems and error measures. We consider the variational models of Section 3 on $\Omega = (0,1)^d$, with $d = 1, 2$. Smooth exact solutions satisfying homogeneous natural boundary conditions are prescribed, and the corresponding loads are determined from the governing equations. Table 1 summarizes the model parameters and activation degrees. For the pure and regularized p-energies, the approximations satisfy the zero-mean constraint.

For a raw ReLU^k ridge function φ, let $P_\circ$ denote mean removal in the pure and regularized p-models and the identity otherwise. With Q_{tr} denoting the quadrature used to evaluate the energy and its derivative, the computational atoms are

$$g_Q = \frac{P_\circ \varphi}{\mathcal{N}_Q(P_\circ \varphi)},$$

TABLE 1
Model problems and parameters.

Energy	d	p	k
Linear reaction–diffusion	1	—	3
Cubic semilinear	1	—	3
Hyperbolic-sine semilinear	2	—	3
Pure p-Laplacian	1	4	3
Pure p-Laplacian	2	4	3
Regularized p-Laplacian, $\varepsilon = 0.1$	2	4	3
Reaction p-Laplacian	2	4	3
Pure p-Laplacian	1	5	1

where

$$\mathcal{N}_Q(v) = \begin{cases} \left[Q_{\mathrm{tr}}(|v|^2 + |\nabla v|^2)\right]^{1/2}, & \text{linear and semilinear models,} \\ \left[Q_{\mathrm{tr}}(\sum_{i=1}^d |\partial_i v|^p)\right]^{1/p}, & \text{pure and regularized } p\text{-models,} \\ \left[Q_{\mathrm{tr}}(|v|^p + \sum_{i=1}^d |\partial_i v|^p)\right]^{1/p}, & \text{reaction } p\text{-model.} \end{cases}$$

For the p-models, the normalization and reported Sobolev errors use the componentwise gradient convention above. The energies and the natural V-map use the Euclidean gradient norm.

The finite-pool CGA constructs nested approximation spaces by selecting linearly independent atoms according to the evaluated derivative score and reoptimizing all coefficients in each enlarged space. The approximation dimension N is the number of selected independent atoms, equivalently the dimension of the current trial space. In this section, the error curves are sampled at dyadic dimensions $N = 2^j$, which are also used to compute the local convergence orders. The prescribed solutions and further implementation details are given in the supplementary material.

For the linear and semilinear cases we report the energy gap and

$$e_{H^1}(\widehat{G}_N) = \frac{\left\|\widehat{G}_N - u^*\right\|_{H^1}}{\|u^*\|_{H^1}}.$$

For the pure and regularized p-models we report the relative gradient seminorm, with the componentwise convention

$$|v|_{1,p} = \left(\int_\Omega \sum_{i=1}^d |\partial_i v|^p \,\mathrm{d}x\right)^{1/p}, \qquad e_{\nabla,p}(\widehat{G}_N) = \frac{|\widehat{G}_N - u^*|_{1,p}}{|u^*|_{1,p}}.$$

For the reaction p-model, the reported relative error uses the full norm $\|v\|_{W^{1,p}} = (\|v\|_{L^p}^p + |v|_{1,p}^p)^{1/p}$. On the zero-mean space, the gradient seminorm and full Sobolev norm are equivalent and have the same theoretical error powers. Each relative metric uses its own denominator. The relative natural distance is $e_V(\widehat{G}_N) = d_{V,\bullet}(\widehat{G}_N, u^*)/d_{V,\bullet}(0, u^*)$, with a nonzero denominator for the benchmark profiles. For evaluation quadrature Q_{ev}, write

$$\widehat{\delta}_{N,Q_{\mathrm{ev}}} = \mathcal{E}_{Q_{\mathrm{ev}}}(\widehat{G}_N) - \mathcal{E}_{Q_{\mathrm{ev}}}(u^*),$$

where $\mathcal{E}_{Q_{\mathrm{ev}}}$ denotes the quadrature-evaluated energy. The numerical records retain signed energy differences, and energy summaries use the metric-valid observations defined in Section 6. The dyadic local order is

$$\mathrm{ord}_N(e) = \log_2\left(\frac{e(N/2)}{e(N)}\right).$$

The figures and tables report energy gaps, relative Sobolev errors, and relative natural distances, together with the corresponding dyadic local orders.

5.2. Reference powers and metric conversion. For ReLU^k atoms, the Hilbert-space approximation exponent associated with the smooth-ridge entropy estimate is

$$\beta_H = \frac{1}{2} + \frac{2(k-1)+1}{2d}.$$

The dashed segments show the reference powers $N^{-2\beta_H}$ for the energy gap, $N^{-\beta_H}$ for the H^1 error or natural V-distance, and $N^{-2\beta_H/p}$ for the Sobolev error of a standard p-structure energy. These powers follow the conversion in Corollary 3.7. Their exponents are prescribed, and the vertical offsets are set by the median log error over the last up to three positive dyadic states.

5.3. Linear and cubic calibration. Figure 1 shows the retained dyadic states for the linear and cubic multifrequency problems. Both curves decrease rapidly after about 32 atoms. Between $N = 128$ and 256, the linear energy and H^1 orders are 6.083 and 3.042, whereas the cubic orders are 6.896 and 3.448; the corresponding terminal relative H^1 errors are 1.247×10^{-4} and 1.215×10^{-4}.

For the two-dimensional sinh model, the relative H^1 error decreases from 4.074×10^{-3} at $N = 256$ to 1.990×10^{-3} at $N = 512$. The local orders are 1.793 on $128 \to 256$ and 1.034 on $256 \to 512$; the trajectory is shown in the supplementary figures.

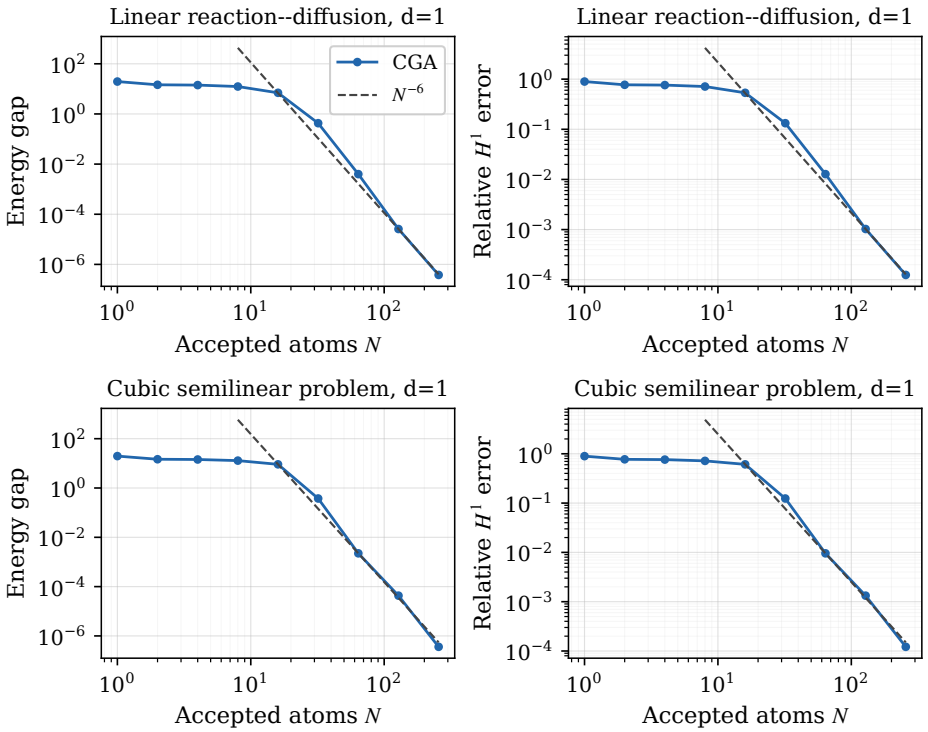


FIG. 1. *CGA errors at dyadic retained widths for the linear and cubic problems. The dashed segments show the reference powers N^{-6} for the energy gap and N^{-3} for the relative H^1 error.*

5.4. Numerical results for the pure and regularized $p = 4$ models. Figure 2 reports the energy gap, relative gradient L^4 seminorm error, and relative V-distance for the one- and two-dimensional pure $p = 4$ models. At $N = 128$, the one-dimensional case has relative gradient and V-errors of 1.579×10^{-5} and 1.680×10^{-6}. For the two-dimensional case, the energy and V-orders on $128 \to 256$ are 3.410 and 1.703. From $N = 256$ to 512, the V-distance decreases, while the relative gradient errors are 2.494×10^{-3} and 3.601×10^{-3}, respectively.

The dyadic values for the semilinear and $p = 4$ cases are supplied in the supplementary numerical records. The regularized and reaction trajectories show the same separation between the natural-distance and Sobolev metrics as the representative pure cases.

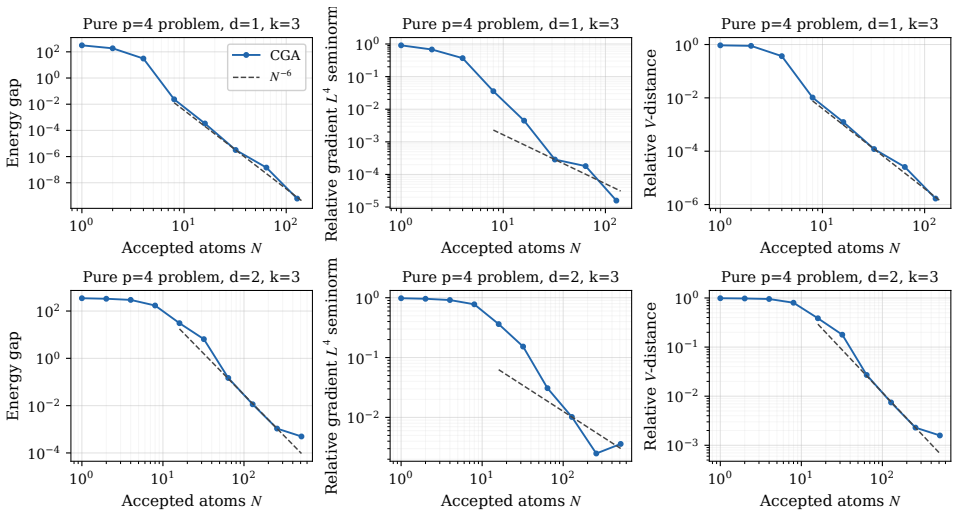


FIG. 2. *CGA errors at dyadic retained widths for the pure $p = 4$ models. Columns show the energy gap, relative gradient L^4 seminorm error, and relative V-distance. Dashed segments show powers $N^{-6}, N^{-3/2}, N^{-3}$ in one dimension and $N^{-7/2}, N^{-7/8}, N^{-7/4}$ in two dimensions, in column order.*

5.5. Finite-trajectory error estimates for quartic energies. We apply the projection estimates and quartic energy comparison of Subsection 4.3 to the one-dimensional pure and regularized $p = 4$ models. We take activation degree $k = 3$, exact solution $u^*(x) = \cos(2\pi x)$, and $\varepsilon = 0$ and 0.1, respectively. The approximate source representation is constructed at $N = 8$ and then held fixed. The comparison constants are determined on the states $N = 8, \dots, 32$, while the decrease constant is determined from the corresponding transitions $N \to N+1$, $N = 8, \dots, 31$. Each trajectory is then continued to $N = 64$, with the source representation and these constants unchanged.

The approximate-source estimate bounds the squared projection error r_N^2, and the quartic comparison transfers this estimate to the energy error of the computed states. Throughout the calibration range, the resulting energy bounds cover the computed energy errors for both models.

To examine the power-type decrease condition, we prescribe the benchmark exponent $\alpha_{\mathrm{bench}} = 6$ and define

$$U_j = \left(r_8^{-2/\alpha_{\mathrm{bench}}} + \frac{c_0 j}{\alpha_{\mathrm{bench}}} \right)^{-\alpha_{\mathrm{bench}}}, \qquad 1 \le j \le 24.$$

Under the stepwise decrease condition

$$\gamma_N^2 \ge c_0 r_N^{2+2/\alpha_{\mathrm{bench}}},$$

the finite-range estimate in Theorem 4.10 gives $r_{8+j}^2 \le U_j$. Applying Corollary 4.11 yields energy, natural-distance, and $W^{1,4}$-error bounds proportional to U_j, $U_j^{1/2}$, and $U_j^{1/4}$, respectively.

During continuation, the same decrease condition is tested with the calibrated value of c_0. Among the 32 transitions $N \to N+1$, $N = 32, \dots, 63$, the condition holds for 27 transitions in the pure model and 28 in the regularized model. The source

quantities, comparison constants, and quadrature consistency checks are provided in the supplementary material.

6. Equal-DOF baseline comparisons.

6.1. Comparison setup. To assess approximation accuracy at matched coefficient counts, we compare five cases using a common evaluator: linear and cubic energies in one dimension, the sinh energy in two dimensions, and pure $p = 4$ energies in one and two dimensions, denoted C1–C5 in this order. The comparison counts retained CGA/RFM coefficients and independent FEM coefficients after removal of the constant mode where required.

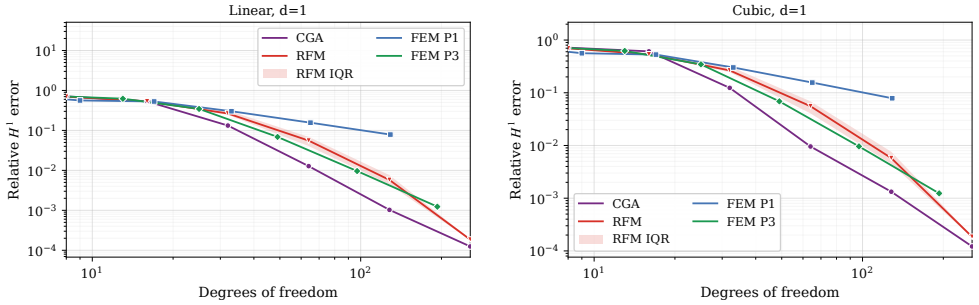
For CGA, the comparison optimizes coefficients on fixed greedy prefixes. FEM uses uniform one-dimensional or structured triangular meshes with P1 and P3 references and supplementary P2 states. RFM uses nested normalized ReLU^3 features with ten prescribed seeds, 201–210, at each width. Supplement S4 specifies the sampling, fitting, and evaluation protocol.

Figures join computed dyadic states and terminal CGA prefixes, while FEM curves use the mesh coefficient counts. The display ranges end at the corresponding CGA terminal widths. Table endpoints are $256, 256, 256, 128, 256$ for C1–C5. FEM values use a computed state at the requested count when available and otherwise log–log interpolation between bracketing metric-valid states. The C3 P3 value is computed directly at $N = 256$.

6.2. Evaluation and statistical samples. An observation enters a metric summary when the solve succeeds, the metric is finite and nonnegative, and successive evaluation quadratures have a relative difference of at most one percent. The same rule applies to all methods. Energy differences retain their sign in the numerical records. Each of the energy, Sobolev, and V-distance summaries uses its own valid observations.

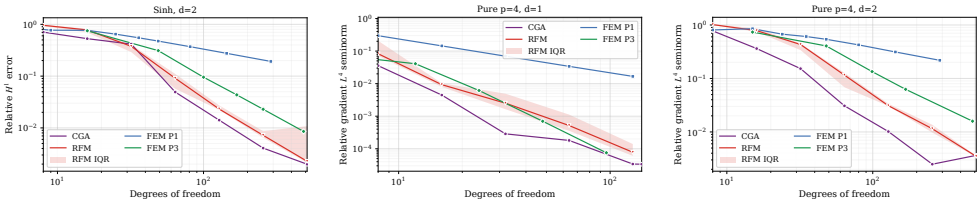
RFM medians and interquartile ranges are computed from the valid observations for each metric. In Table 2, *Valid* gives the number of valid Sobolev observations out of the ten prescribed seeds. Supplement S3 lists the solve statuses, metric sample counts, and model identifiers.

6.3. Relative Sobolev errors. C1–C3 report relative H^1 errors, and C4–C5 report the relative componentwise gradient L^4 seminorm $e_{\nabla,4}$ defined in Section 5. The V-distance is reported in the following subsection.



(a) Linear, $d = 1$. (b) Cubic, $d = 1$.

FIG. 3. *Relative H^1 error versus coefficient count for the one-dimensional comparison. RFM bands show metric-valid medians and interquartile ranges from ten prescribed seeds. Markers are computed states; CGA coefficients are refitted on fixed greedy prefixes.*



(a) Sinh, $d = 2$: H^1. (b) Pure $p = 4, d = 1$: $e_{\nabla,4}$. (c) Pure $p = 4, d = 2$: $e_{\nabla,4}$.

FIG. 4. *Relative H^1 error for sinh and relative gradient L^4 seminorm for the pure $p = 4$ problems. RFM bands summarize the metric-valid observations from ten prescribed seeds. Endpoint valid counts appear in Table 2.*

Table 2 reports the endpoint errors and comparator-to-CGA ratios. A ratio above one indicates a smaller CGA error. CGA has smaller errors than the metric-valid RFM medians at all five endpoints and smaller errors than the P3 FEM references in four cases. For the one-dimensional pure $p = 4$ case at $N = 128$, the P3/CGA and RFM-median/CGA ratios are approximately 0.88 and 2.30, respectively.

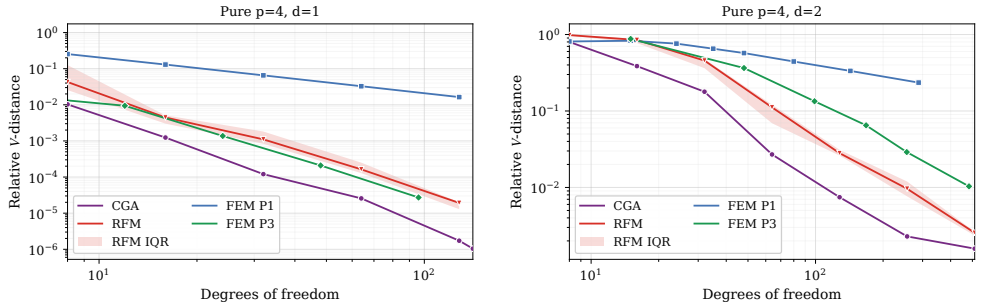
TABLE 2

Endpoint relative H^1 errors for C1–C3 and relative gradient L^4 seminorm errors for C4–C5. RFM values are median [Q1,Q3] over metric-valid observations from ten prescribed seeds. Valid gives the Sobolev sample count; the inclusion rule is defined in the text. The P3 value for C3 is computed directly, and the other P3 endpoint values use log–log interpolation between bracketing metric-valid states. Ratios above one indicate smaller CGA errors.

Problem	DOF	CGA	RFM median [Q1,Q3]	Valid	RFM/CGA	FEM P3/CGA
Linear, d=1	256	1.247×10^{-4}	1.853×10^{-4} [1.779×10^{-4}, 1.913×10^{-4}]	10/10	1.49	4.24
Cubic, d=1	256	1.215×10^{-4}	1.853×10^{-4} [1.779×10^{-4}, 1.913×10^{-4}]	10/10	1.52	4.35
Sinh, d=2	256	4.074×10^{-3}	7.196×10^{-3} [6.472×10^{-3}, 8.663×10^{-3}]	10/10	1.77	5.63
Pure p=4, d=1	128	3.395×10^{-5}	7.818×10^{-5} [6.046×10^{-5}, 1.400×10^{-4}]	10/10	2.30	0.88
Pure p=4, d=2	256	2.493×10^{-3}	1.147×10^{-2} [9.730×10^{-3}, 1.360×10^{-2}]	6/10	4.60	14.54

6.4. Natural V-distance for the p-models. Figure 5 reports the relative V-distance for the pure $p = 4$ models. Each curve uses computed states, and the RFM bands show the median and interquartile range of the observations valid for this

metric.



(a) Pure $p = 4, d = 1$. (b) Pure $p = 4, d = 2$.

FIG. 5. *Relative V-distance versus coefficient count for the pure $p = 4$ comparisons. Markers show computed states; RFM bands show metric-valid medians and interquartile ranges.*

7. Discussion and conclusion. This work develops a variational analysis of finite-pool Chebyshev greedy approximation for nonlinear PDE energies. Under the atomic source condition, directional smoothness, and uniform positive selection strength, the ideal continuous CGA has an $O(m^{-1})$ energy bound. For finite-pool CGA, the one-step estimate quantifies the contributions of dictionary coverage, score evaluation, radial stationarity, and coefficient optimization. Relative accuracy conditions preserve the same energy order.

For the pure, regularized, and reaction p-energies with $p \geq 2$, the energy gap controls both the natural V-distance and the Sobolev error. An energy exponent α yields the metric exponents $\alpha/2$ and α/p, respectively. These relations translate greedy energy estimates into error measures adapted to the nonlinear PDE geometry.

Local Hessian transfer and finite-trajectory projection estimates connect the descent analysis with approximation theory. The Hessian argument relates nonlinear selection to weak OGA in a frozen Hilbert metric. The finite-trajectory argument uses approximate source representations and normalized innovations to bound errors in the generated nested spaces. For quartic energies, the exact expansion converts these projection estimates into nonlinear energy bounds.

The one-dimensional pure and regularized $p = 4$ experiments illustrate the finite-trajectory estimates. On $N = 8, \ldots, 32$, the source and quartic comparison bounds cover the computed energy errors. The prescribed benchmark $\alpha_{\mathrm{bench}} = 6$ gives comparison powers of 6, 3, and 3/2 for energy, natural distance, and Sobolev error. With the calibration constants fixed, the normalized-innovation condition holds on 27 of 32 continuation transitions for the pure model and 28 of 32 for the regularized model.

At the five prescribed comparison endpoints, CGA has smaller Sobolev errors than the metric-valid RFM medians and smaller errors than the P3 FEM references in four cases. These comparisons quantify the approximation accuracy obtained by adaptive greedy selection at matched coefficient counts.

The finite-pool implementation requires a balance between directional coverage and the costs of candidate evaluation and repeated coefficient optimization over growing trial spaces. These computational demands motivate residual-guided pool enrichment, preconditioned coefficient solvers, and adaptive optimization tolerances informed by

the inexact-descent estimates. On the theoretical side, further questions concern the uniform control of the local Hessian-transfer and quartic-comparison conditions as the approximation spaces grow, and the construction of error estimators that can be evaluated without knowledge of the exact solution. The present numerical study focuses on smooth manufactured solutions in one and two dimensions. Extending these experiments to higher-dimensional problems and solutions of limited regularity, together with systematic comparisons at matched computational budgets, would clarify the scalability of CGA and the relation between its approximation accuracy and computational cost.

Together, the descent estimates, natural-distance relations, and projection bounds provide a quantitative framework for adaptive ridge approximation of nonlinear PDE energies. The numerical results demonstrate how finite-pool greedy selection constructs accurate trial spaces for the representative linear, semilinear, and p-structure problems.

8. Code and data availability. Code, configurations, and numerical data are available at https://github.com/RonzolYu/cga4PDE. The repository contains the coefficient models, evaluation rules, and records underlying the figures and tables. Supplements S1–S4 describe the experiment configurations, source construction, statistical samples, and reproduction commands.

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